MTH1001

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

January 2020

Mathematical Structures

Module Leader: Prof. Nigel Byott

Other Examiner: Dr Gihan Marasingha

Duration: 2 HOURS.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) In this question, select **one or more** possible deductions from a list of premises. You need not provide reasoning.

   (i) Given the premises P and Q, which of the following can be deduced: $P \wedge Q$, $P \vee Q$, $\neg P$, $\neg Q$, $\neg(P \wedge Q)$?

   (ii) Given the premises $P \vee Q$ and $P \vee \neg Q$, which of the following can be deduced: $P \wedge Q$, P, Q, $\neg P$, $\neg Q$?

   (iii) Given no premises, which of the following can be deduced: P, Q, $Q \wedge \neg Q$, $P \vee \neg P$, $\bot$?

   (iv) Given the premise $\bot$, which of the following can be deduced: $\bot$, P, Q, $\neg P$, $\neg Q$? (4)

   (b) Identify the free and bound occurrences of variables in the following expressions.

   (i) j^3, (ii) $\sum_{j=0}^{n} j^3$, (iii) $(x > y) \wedge (\exists x\ x > 10)$,

   (iv) $\forall x\ (x > y)$, (v) $\exists y\ \forall x\ (x > y)$.

   (10)

   (c) Let $A = \{2, 3, 5\}$, $B = \{2, 3, 8\}$, $C = \{3, 5, 7, 8\}$, $D = \{x, y\}$ be sets. Express each of the following sets as a list of elements delimited by braces.

   (i) $A \cup B$, (ii) $A \cap C$, (iii) $C \setminus A$,
   (iv) $\mathcal{P}(D)$, (v) $A \times D$, (vi) $A \cap (B \cap C)$.

   (6)

   (d) Let A and B be sets.

   (i) Define what it means for R to be a relation on A and B.

   (ii) Define what it means for R to be a relation on A.

   (iii) Define what it means for a relation R on a set A to be symmetric.

   (iv) Let S be the relation on the set $\mathbf{Z}$ of integers where $x\ S\ y$ is defined to mean 'y is a multiple of x'. Is S symmetric? Give a proof or a counterexample. (10)

(e) Let $z = 1 + i$ and let $w = \sqrt{2}(\cos\alpha + i\sin\alpha)$, where α is a real number.

(i) Write z in polar form.

(ii) Write zw in polar form.

(iii) Let n be an integer. Write $(zw)^{8n}$ in polar form, simplifying as much as possible. (10)

(f) (i) Let (a_n) be a sequence of complex numbers and let a be a complex number. Define what it means for (a_n) to converge to the limit a.

(ii) Let (s_n) and (t_n) be convergent sequences of complex numbers with limits s and t, respectively. Show that the sequence $(s_n + t_n)$ converges to the limit $s + t$. (10)

[50]

SECTION B

2. (a) Using rules of inference for the propositional calculus,

   (i) deduce $P \wedge (Q \vee R)$ from the premises P and Q;

   (ii) deduce S from the premise $(S \vee T) \wedge (T \implies S)$.

   You need not explicitly name each rule of inference. (10)

   (b) (i) State the rule of mathematical induction.

   (ii) Let (a_n) be a sequence of real numbers defined inductively by $a_1 = 1$ and $a_{n+1} = \sqrt{(6 + a_n^2)/3}$, for every natural number n. By induction, prove that $a_n^2 < 3$, for every natural number n.

   (iii) Prove that (a_n) is convergent and find its limit. (15)

   [25]

3. (a) Let A, B, C be sets. Let $f : A \to B$ and $g : B \to C$ be functions.

   (i) Give the definition of the composite $g \circ f$.

   (ii) Define what it means for f to be injective.

   (iii) If f and g are injective, does it follow that $g \circ f$ is injective? Give a proof or a counterexample.

   (iv) If $g \circ f$ is injective, does it follow that f and g are injective? Give a proof or a counterexample. (13)

   (b) (i) Let (a_n) be a sequence of complex numbers. Define what it means for a series

   $$\sum_{n=1}^{\infty} a_n$$

   to converge.

   (ii) If $\sum_{n=1}^{\infty} a_n$ converges, does it follow that $a_n \to 0$ as $n \to \infty$? Give a proof or a counterexample.

   (iii) If $a_n \to 0$ as $n \to \infty$, does it follow that $\sum_{n=1}^{\infty} a_n$ converges? Give a proof or a counterexample. (12)

   [25]

4. (a) Let A and B be sets.

   (i) Define what it means for A to be a subset of B, written $A \subseteq B$.

   (ii) Define what it means for A to equal B.

   (iii) Give the definition of the union, written $A \cup B$, and the intersection, written $A \cap B$, of A and B.

   (iv) Give the definition of the set difference $A \setminus B$.

   (v) Prove that $(A \cap B) \cup (A \setminus B) = A$, for any sets A and B. (14)

   (b) Let S be a subset of $\mathbf{R}$, the set of real numbers.

   (i) Define what it means for $x \in \mathbf{R}$ to be an upper bound of S.

   (ii) Define what it means for $z \in \mathbf{R}$ be to be a supremum of S.

   (iii) Prove that for any subset S of $\mathbf{R}$, if u is a supremum of S and v is a supremum of S, then $u = v$. (11)

[25]